

Task for Angular .Net Full stack Developer :

Design a system for an Industrial Manufacturing setup having 50 machines and 200 types of parts being produced.

1. Machine Master : This will have the following properties:
 - a) Id (int64)
 - b) Machine Name (varchar(64))
 - c) Machine Serial Number (varchar(16))
 - d) Machine Manufacturer Id (int64)
 - e) Machine Model (varchar(16))
 - f) Year of Manufacture (int16)
 - g) Type of Machine (int32)
 - h) Location Id (int64)
2. For the above properties, Machine Manufacturer Id will come from a separate master called Machine Manufacturer Master which will have the following properties:
 - a) Id (int64)
 - b) Name (varchar(128))
3. Location Id will come from a master called Location Master with the following properties:
 - a) Id (int64)
 - b) Name (varchar(128))
 - c) Latitude (decimal(8, 6))
 - d) Longitude (decimal(9, 6))
4. Type of Machine will come from an enum which will contain the following members :
 - a) CNC_Turning_Center = 1
 - b) VMC = 2
 - c) HMC = 3
 - d) HBM = 4
 - e) VTL = 5
 - f) 5-Axis-Machining-Center = 6
5. Component Master : This will have the following properties :
 - a) Id (int64)
 - b) Customer Id (int64)
 - c) Component Name (varchar(64))
 - d) PartNo (varchar(32))
 - e) ECN (varchar(16))
6. The Customer Id from the above will come from a master called Customer Master with the following properties :
 - a) Id
 - b) Name

7. Component Operation Master :
 - a) Component Id (int64)
 - b) Machine Id (int64)
 - c) Operation Code (varchar(16))
 - d) Operation Name (varchar(64))
 - e) Operation Description (varchar(MAX))
 - f) Operation Type (int32)

8. Operation Type will come from an enum with the following members :
 - a) Turning = 1
 - b) Milling = 2
 - c) Drilling = 3
 - d) Chamfering = 4
 - e) Tapping = 5
 - f) Threading = 6
 - g) Boring = 7
 - h) Knurling = 8
 - i) Honing = 9
 - j) Buffing = 10

Generate the classes and database structure for the above requirements in ASP.Net Core Web API 3.1 or above. Make a controller to accept requests and save the incoming data into the database.

In the angular application, make a master and detail component for every type of entity defined above. In the master view, there should be a table / data grid displaying all the components currently saved in the database. In the detail view, if there is an association between entities like for Machine or Component Operation, the user should be able to select from a dropdown.

The angular application should call the Web API defined above to get all the required data. Set up angular routes correctly and make the application work correctly.